EMTCAL: Efficient Multiscale Transformer and Cross-Level Attention Learning for Remote Sensing Scene Classification

Xu Tang, *Senior Member, IEEE*, Mingteng Li, Jingjing Ma, *Member, IEEE*, Xiangrong Zhang, *Senior Member, IEEE*, Fang Liu, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— In recent years, convolutional neural network (CNN)-based methods have been widely used for remote sensing (RS) scene classification tasks and have achieved excellent results. However, CNNs are not good at exploring contextual information, which is essential for fully understanding RS scenes. A new model named transformer attracts researchers' attention to address this problem, which is skilled in mining the latent contextual information in RS scenes. Nevertheless, since the contents of RS scenes are diverse in type and various in scale, the performance of the original transformer in RS scene classification cannot reach what we expect. In addition, due to the specific self-attention mechanism, the time costs of the transformer are high, which hinders its practicability in the RS community. To overcome the above limitations, we propose a new model named efficient multiscale transformer and cross-level attention learning (EMTCAL) for RS scene classification in this article. EMTCAL combines the advantages of CNN and transformer to mine information within RS scenes fully. First, it uses a multilayer feature extraction module (MFEM) to acquire global visual features and multilevel convolutional features from RS scenes. Second, a contextual information extraction module (CIEM) is proposed to capture rich contextual information from multilevel features. In CIEM, taking the characteristics of RS scenes and the computational complexity into account, we propose an efficient multiscale transformer (EMST). EMST can mine the abundant knowledge with various scales hidden in RS scenes and model their inherent relations at small time costs. Third, a cross-level attention module (CLAM) is developed to aggregate and explore correlations of multilevel features. Finally, a class score fusion module (CSFM) is designed to integrate the contributions of global and aggregated multilevel features for discriminative scene representations. Extensive experiments are conducted on three public RS scene datasets. The positive results demonstrate that our EMTCAL can achieve superior classification performance and outperform many state-of-the-art methods. Our source codes are available in https://github.com/TangXu-Group/Remote-Sensing-Images-Classification/tree/main/EMTCAL.**

***Index Terms*— Cross-level attention, remote sensing (RS) scene classification, transformer.**

I. Introduction

REMOTE sensing (RS) scene classification is a foundational task in the RS community, which can be used in many practical applications, such as land cover classification, geological survey, urban planning, and hazard monitoring [1]–[4]. Its goal is to classify RS scenes into different semantic groups according to their contents. In the past decades, extensive methods have been proposed for RS scene classification [5], [6]. One of their key points is feature extraction/learning. In the beginning, the methods mainly relied on handcrafted features, such as histogram of oriented gradients (HOG) [7] and scale-invariant feature transform (SIFT) [8]. However, their performance is not satisfactory since the handcrafted features cannot fully capture the complex contents of RS scenes. Recently, with the appearance of convolutional neural networks (CNNs), an ocean of CNN-based RS scene classification methods have been proposed [9]–[15]. Due to the data-driven learning manner and hierarchical structure, CNNs can automatically learn rich semantic information from RS scenes. Therefore, CNN-based methods achieve remarkable performance, dominating the RS scene classification task.

Although CNN-based methods have significantly improved RS scene classification performance, they are not good at exploring the long-range contextual information hidden in RS scenes, which is essential to interpret RS scenes fully. For instance, as shown in Fig. 1, CNNs focus on extracting features from local regions using convolutional operators but ignore the long-range contextual dependencies between local regions [16]–[18]. We can find that the "Airport" scene (see Fig. 1) contains many scene-related ground objects, such as "Airplane," "Runway," and "Building." Due to high interclass similarity in RS scenes, if the models only focus on learning local structure information, the "Airport" scene may be misclassified as other similar scenes (e.g., "Airplane"). Therefore, the context relationships of various

Manuscript received 29 May 2022; revised 1 July 2022; accepted 25 July 2022. Date of publication 28 July 2022; date of current version 9 August 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62171332, in part by the Key Research and Development Program of Shaanxi under Grant 2021GY-035 and Grant 2019ZDLGY03-08, in part by the Fund of National Key Laboratory of Science and Technology on Remote Sensing Information and imagery Analysis, in part by the Beijing Research Institute of Uranium Geology under Grant 6142A010301, and in part by the Open Research Fund in 2021 of Jiangsu Key Laboratory of Spectral Imaging Intelligent Sense under Grant JSGP202101. *(Corresponding author: Jingjing Ma.)*

Xu Tang, Mingteng Li, Jingjing Ma, Xiangrong Zhang, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: tangxu128@gmail.com; jjma@ieee.org).

Fang Liu is with the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China.

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2022.3194505, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2022.3194505

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

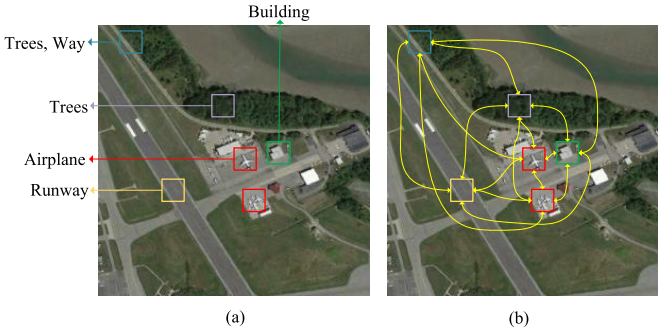


Fig. 1. (a) Local feature and (b) long-range contextual information within a scene selected from the "Airport" class. CNNs are good at extracting local features but overlook their long-range contextual information. From observing (a), we can find this scene contains many distinct land covers, such as "Airplanem" "Runway," "Building," and "Tree." If the models only focus on learning those local structural features, the "Airport" scene may be misclassified as other scenes (e.g., "Airplane") with similar ground objects. Therefore, the expected classification models should consider local land covers and their contextual relationships to make correct decision, especially long-range connections. Taking (b) as an example, when the relationships between "Airplane," "Runway," and "Building" can be adequately interpreted, this scene can be classified into "Airport" correctly.

local information should be considered entirely. Scholars always make attempts to address the problems discussed above, such as the nonlocal approach [19] and the conformer strategy [20]. Among the various solutions, transformer [21], [22] draws the scholars' attention because of the effectiveness of learning contextual information. Lately, many successful transformer-based models have been proposed for diverse computer vision tasks [23]–[26].

However, the transformer model still has some room for improvement in interpreting RS scenes. First, the traditional transformer uses image patches as input and models the contextual dependencies between patches, which has limitations for exploring local structural information of RS scenes with complex spatial patterns and geometric structures. Second, due to transformer's specific operation mechanism that captures the contextual information by measuring relationships between all the image patches, the original transformer has high computational complexity [27]. Third, the original transformer can only extract contextual information at a single scale without considering the multiscale property of RS scenes. In addition, most RS scene classification methods use global visual features to classify RS scenes while ignoring multilevel local information, which is also crucial for understanding RS scenes. Based on the above discussion, three questions will arise naturally: 1) whether the advantages of CNNs and transformer can be combined or not for the exploration of local information and contextual relationships; 2) can we design an efficient RS-oriented transformer for RS scene classification task; and 3) how to efficiently aggregate global visual feature and multilevel local information for generating discriminative feature representations.

Based on the above discussion, we propose a new method for RS scene classification in this article, named efficient multiscale transformer and cross-level attention learning (EMTCAL). First, we use a multilayer feature extraction module (MFEM) to extract global visual features and multilevel convolutional features from RS scenes. Second, a contextual information extraction module (CIEM) with efficient multiscale transformers (EMSTs) is proposed to capture contextual information hidden in RS scenes comprehensively and thoroughly. By embedding dilated convolution [28] and adjusting the self-attention manner, not only can multiscale information within RS scenes be mined but also computational consumption can be reduced. Accordingly, it is more suitable for RS scene classification. Third, a cross-level attention module (CLAM) is developed to model correlations of different level features and aggregate multilevel features. Thus, the global contextual understanding capacity of our EMTCAL can be enhanced. Finally, we design a simple yet effective class score fusion module (CSFM) to simultaneously integrate the contributions of aggregated local and global visual features.

Many techniques can explore the contextual information hidden in RS scenes, such as graph convolutional network (GCN) [29]. In this article, we develop an EMST to explore the intrinsic contextual knowledge within RS scenes. Here, the reasons why we select the transformer rather than other models are twofold and summarized as follows. First, the transformer is a model that relies on the self-attention mechanism to explore long-range dependencies within the input data [21]. It comprehensively captures the contextual information hidden in RS scenes by creating the relationships between all the pixels. In contrast, GCNs focus on aggregating adjacent nodes to obtain contextual clues, limiting the obtained materials' integrity. Second, the core part of the transformer is the multihead self-attention (MSA). Multiheads are able to extract contextual information within RS scenes from different representative subspaces so that the transformer has a strong feature representation capacity. Here, each head of MSA can be regarded as a special GCN or other similar models [30].

The major contributions of this article are summarized as follows.

1) We propose an RS scene classification network EMTCAL. On one hand, it can effectively combine CNN and transformer's advantages to extract local information and contextual correlations. On the other hand, both multilevel local information and global visual features within RS scenes can be thoroughly mined and aggregated.
2) The EMST is proposed. Compared with the original transformer models, it can mine multiscale information and contextual information from RS scenes effectively and efficiently. Correspondingly, it is more robust and suitable for RS scene classification.
3) A CLAM is proposed to simultaneously model correlations of different level features and aggregate multilevel features, which is beneficial for understanding RS scenes adequately.
4) Extensive experiments are conducted on three benchmark datasets, and the encouraging results demonstrate that the proposed EMTCAL is effective in RS scene classification.

The rest of this article is organized as follows. Section II briefly reviews the literature related to CNN-based methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

and attention-based methods for RS scene classification. Then, the proposed method is introduced in Section III. The experimental results and discussion are provided in Section IV. Finally, we draw a conclusion in Section V.

II. Related Work

A. CNN-Based Methods for RS Scene Classification

We roughly divide the existing CNN-based methods into three groups. The methods of the first group focus on learning global visual features for RS scene classification. For example, Li *et al.* [31] combined transfer learning and CNN to learn a robust visual feature representation and improve the classification accuracy with limited labeled samples. Meanwhile, to overcome the limitation of fixed input size required by pretrained network, a scale-free CNN was proposed in [32], which can process RS scenes with arbitrary sizes by replacing the fully connected (FC) layer as a convolutional layer. Positive classification results confirm its usefulness. To further improve the classification performance, Zhang *et al.* [33] proposed an attribute-cooperated CNN, which exploits attributes of RS scenes. The obtained attributes are regarded as additional information to extract discriminative visual features for RS scene classification.

The second group models aim to aggregate multilayer features for RS scene classification. Lu *et al.* [34] used a pretrained CNN to extract multilayer convolutional features from RS scenes. Then a supervised feature encoding module was proposed to aggregate multilayer features to improve the classification results. A deep feature aggregation framework driven by graph convolutional network (DFAGCN) was proposed in [35], where global visual features and multilayer context information are adequately captured and fused to improve the classification accuracy. Wang *et al.* [36] developed an enhanced feature pyramid network (EFPN) to extract multilevel features from RS scenes. Then a two-branch deep feature fusion module was designed to aggregate multilevel features. Therefore, complementary advantages of multilevel features can be exploited for RS scene classification tasks.

The approaches in the third group are used to combine CNNs with other techniques to improve the performance of RS scene classification further. In the literature [37], a discriminative CNN was proposed to boost the performance of RS scene classification. By introducing the metric learning mechanism, the captured features are more discriminative. Based on the neural architecture search (NAS) method, a model was proposed in [38], which uses an automatic search strategy to find the most suitable network architecture for RS scene classification and achieves encouraging performance with high computational efficiency.

B. Attention-Based Methods for RS Scene Classification

The CNN-based methods have achieved excellent performance in RS scene classification. However, emphasizing critical information and capturing contextual dependency information in complex RS scenes are still two urgent challenges. To address them, the visual attention technologies enter the RS image processing community, and many attention-based methods have been proposed for RS scene classification. For clarity, we divide them into two groups, i.e., regular-attention- and transformer-based methods.

The regular-attention-based models aim to mine salient regions from RS scenes for classification tasks. Tang *et al.* [39] proposed an attention consistent network (ACNet) for RS scene classification, which combines spatial attention and channel attention to extract essential features from RS scenes. Also, an attention consistent model was designed to unify the salient regions so that the discriminative features can be accurately extracted. Another method, named multibranch local attention network, was introduced in [40], in which a convolutional local attention module is proposed to obtain both channel and spatial attention weights effectively. Thereby, important/redundant information of RS scenes can be automatically emphasized/suppressed.

The models within the second group are constructed based on the transformer. Owing to the capacity of capturing contextual information hidden in images, the transformer first prevails in the natural image processing field [41]–[43]. The first attempt in computer vision might be the Vision Transformer (ViT) [44]. It transforms natural images into sequences of image patches, and then the original transformer is adopted to accomplish classification. The positive results confirm the potential of the transformer in image processing. Based on ViT, a multiscale vision transformer was proposed in [45], which is a dual-branch network to extract multiscale features for image classification. Also, an efficient feature fusion module was designed to integrate contributions from two branches so that more prosperous and more robust features can be learned. To reduce the time costs caused by the self-attention mechanism, Liu *et al.* [46] proposed a Swin transformer for various vision tasks. By calculating self-attention within local windows, the time costs can be receded significantly.

In the RS community, the number of transformer-based models is few. A new backbone network named SpectralFormer was developed in [47] for the hyperspectral image (HSI) classification task. SpectralFormer applies the original transformer to extract contextual information from HSIs and proposes a group-wise spectral embedding module and a cross-layer adaptive fusion module to learn local and cross-layer information. Bazi *et al.* [48] introduced transformer to deal with RS scene classification. It uses the original transformer model to capture contextual information within RS scenes and achieves excellent classification results. A homo-heterogeneous transformer learning framework (HHTL) was presented in [49]. Through simultaneously mining homogeneous and heterogeneous information within RS scenes, a comprehensive and discriminative feature representation is constructed for RS scene classification.

III. Proposed Method

A. Overall Framework

The framework of the proposed EMTCAL is illustrated in Fig. 2, which consists of an MFEM, a CIEM, a CLAM, and a CSFM. MFEM aims to extract global visual features and multilevel convolutional features from RS scenes. The goal of CIEM is to extract contextual information within

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

RS scenes at a different level, while the target of CLAM is to model contextual correlations of cross-level features and aggregate multilevel features. Besides, CSFM focuses on fusing global visual features and aggregated local features to construct discriminative feature representation for RS scene classification. Now, let us discuss them in detail.

B. Multilayer Feature Extraction Module

In MFEM, we adopt ResNet34 [50] as the feature extractor. As a successful CNN model, it consists of four residual blocks and an FC layer. The residual blocks extract local spatial information, while the FC layer is used to generate a global feature vector as the classification score [51]. Different residual blocks can capture multilevel convolutional features. Shallow convolutional features usually contain low-level information (e.g., color, texture, and shape). The semantic information can be obtained in high-level convolutional features. Considering the complex contents of RS scenes, features from different levels contribute to RS scene classification. To obtain more comprehensive feature representations, as shown in Fig. 2, we select the outputs of the last three residual blocks as the low-, mid-, and high-level convolutional features, respectively. For convenience of description, we denote them as $\mathbf{C}_1 \in \mathbb{R}^{H_1 \times W_1 \times C_1}$, $\mathbf{C}_2 \in \mathbb{R}^{H_2 \times W_2 \times C_2}$, and $\mathbf{C}_3 \in \mathbb{R}^{H_3 \times W_3 \times C_3}$. Then, global visual feature $\mathbf{g} \in \mathbb{R}^{C_3}$ is generated by a global average pooling layer.

C. Contextual Information Extraction Module

Multilevel convolutional features $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_3$ contain rich local information of RS scenes. However, they do not represent the long-range contextual information within RS scenes, which is important for the RS scene classification task. To fully understand the contents of RS scenes, we propose CIEM for effectively extracting contextual information, and its core part is the EMST. Since EMST processes the convolutional features in the same way, we take $\mathbf{C}_1$ as an example to introduce it in detail.

As shown in Fig. 2, EMST is composed of an efficient multiscale multihead self-attention (EMSMSA) block and a multilayer perceptron (MLP) block. Also, the layer norm (LN) layer is used before each block, and the residual connection is used after each block. Similar to the original transformer [21], the MLP block consists of two FC layers with a Gaussian error linear unit (GELU) activation. Thus, the output of EMST can be written as follows:

$$\mathbf{C}_1^{\text{EMSMSA}} = \text{EMSMSA}\big(\text{LN}\big(\mathbf{C}_1'\big)\big) + \mathbf{C}_1' \tag{1}$$

$$\mathbf{F}_1 = \text{MLP}\big(\text{LN}\big(\mathbf{C}_1^{\text{EMSMSA}}\big)\big) + \mathbf{C}_1^{\text{EMSMSA}} \tag{2}$$

where $\mathbf{F}_1$ is the encoded scene feature. Note that unlike the original vision transformer [44] which directly maps the input RS scene image into a series of patch embeddings as input, we flatten the convolutional feature map $\mathbf{C}_1 \in \mathbb{R}^{H_1 \times W_1 \times C_1}$ into a 1-D sequence of token embeddings $\mathbf{C}_1' \in \mathbb{R}^{N_1 \times C_1}$ (where $N_1 = H_1 \times W_1$) as input. Therefore, local structural information can be obtained from the convolutional feature. After EMST, we reshape the output 1-D feature $\mathbf{F}_1 \in \mathbb{R}^{N_1 \times C_1}$ into 2-D image space and add with $\mathbf{C}_1$ for generating discriminative feature representation $\mathbf{F}_1' \in \mathbb{R}^{H_1 \times W_1 \times C_1}$. Similarly, middle-level feature representation $\mathbf{F}_2' \in \mathbb{R}^{H_2 \times W_2 \times C_2}$ and high-level feature representation $\mathbf{F}_3' \in \mathbb{R}^{H_3 \times W_3 \times C_3}$ can also be obtained.

The critical component of EMST is EMSMSA. Before explaining EMSMSA in detail, we briefly introduce MSA. As shown in Fig. 3(a), suppose the input token embeddings $\mathbf{X} \in \mathbb{R}^{N \times C}$ (where N and C indicate the number and dimension of tokens, respectively), MSA performs h self-attention head functions in parallel to capture the contextual relationship among token embeddings from different representation subspaces and different positions. Then the outputs from different heads are concatenated and projected to the final values. For input $\mathbf{X}$, the formulation of MSA is

$$\text{MSA}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O \tag{3}$$

where $\mathbf{W}^O$ represents the learned matrices and head_i denotes the ith self-attention head function. The definition of head_i is

$$\text{head}_i = \text{Attention}\big(\mathbf{X}\mathbf{W}_i^Q, \mathbf{X}\mathbf{W}_i^K, \mathbf{X}\mathbf{W}_i^V\big) \tag{4}$$

where $\mathbf{W}_i^Q$, $\mathbf{W}_i^K$, and $\mathbf{W}_i^V \in \mathbb{R}^{C \times C/h}$ are the learned parameter matrices, and $\text{Attention}(\cdot)$ is the self-attention head function. The self-attention head function is actually the scaled dot-product attention mechanism, and its formulation is

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{5}$$

where $\{\mathbf{Q}, \mathbf{K}, \mathbf{V}\}$ indicates an input data, $\sqrt{d_k}$ is a scaling factor, d_k is the dimension of $\mathbf{K}$, and $\text{softmax}(\cdot)$ represents the softmax function to generate attention scores.

MSA only learns the information from RS scenes at a single scale. However, since the RS scenes contain diverse and abundant objects, their complex contents cannot be fully interpreted in this way. To overcome this limitation, we propose EMSMSA with the consideration of the multiscale property of RS scenes, and its architecture is shown in Fig. 3(b). First, the input $\mathbf{X} \in \mathbb{R}^{N \times C}$ is reshaped into the 2-D space $\mathbf{X}' \in \mathbb{R}^{H \times W \times C}$, where $H \times W = N$. Similar to MSA, EMSMSA performs h attention head functions to process $\mathbf{X}'$, and the input of h attention heads is denoted as $\{\mathbf{X}_i'\}_{i=1}^h$, where $\mathbf{X}_i' \in \mathbb{R}^{H \times W \times C/h}$. Next, different from MSA using h attention heads to extract information with uniform scale, h attention heads of EMSMSA use dilated convolution [28] with different dilation rates to extract multiscale information from $\mathbf{X}'$. Compared with traditional convolution, dilated convolution can expand the receptive field without increasing parameters. Thereby, the features extracted by h attention heads have pyramidal receptive fields. Specifically, for $\mathbf{X}_i'$, this step can be represented as

$$\mathbf{D}_i = \text{DilatedConv}_i\big(\mathbf{X}_i'\big) \tag{6}$$

where $\mathbf{D}_i$ is the generated feature by dilated convolution, and $\text{DilatedConv}_i(\cdot)$ denotes the dilated convolution function of the ith head. Then, we introduce conditional position encoding [52] into our EMSMSA to retain positional information. Unlike fixed or learnable positional encodings in original transformer [44], conditional positional encoding can keep the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

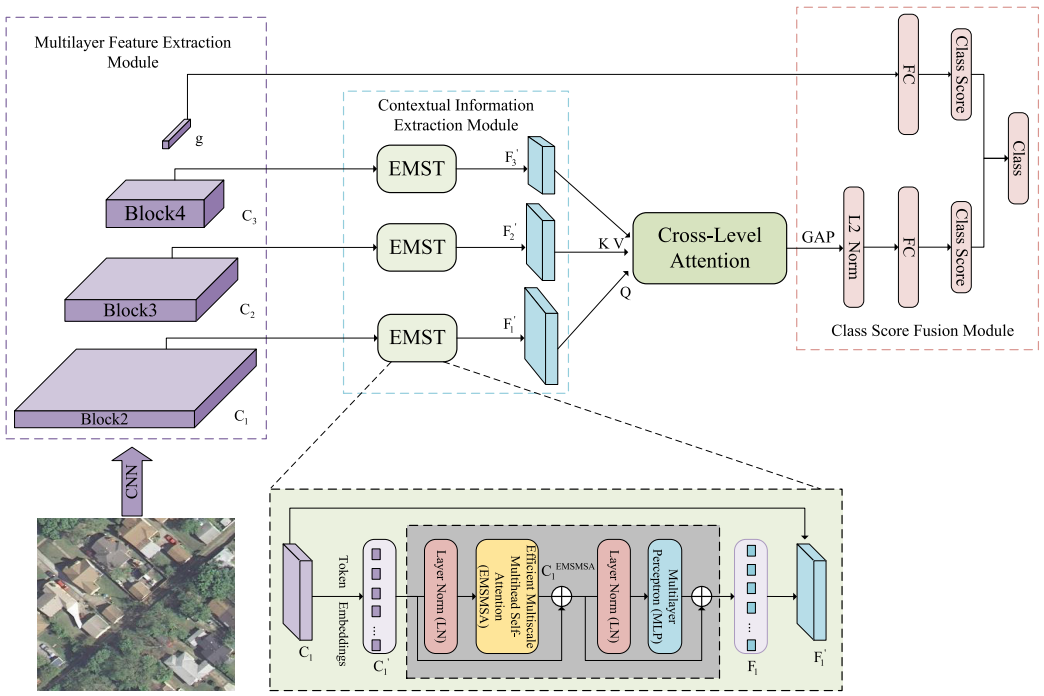


Fig. 2. Framework of the proposed EMTCAL, which consists of an MFEM, a CIEM, a CLAM, and a CSFM.

desired translation invariance for classification task. Conditional positional encoding can be efficiently implemented with a depth-wise (DW) convolution with zero paddings, which can be formulated as

$$\mathbf{D}_i^{\mathrm{CPE}} = \mathrm{DWConv}(\mathbf{D}_i) + \mathbf{D}_i \tag{7}$$

where $\mathrm{DWConv}(\cdot)$ indicates the DW convolution with kernel size 3×3, and $\mathbf{D}_i^{\mathrm{CPE}}$ is the generated feature. Next, we flatten and concatenate these features

$$\mathbf{D} = \mathrm{LayerNorm}\left(\mathrm{Concat}\left(\mathbf{D}_1^{\mathrm{CPE}}, \mathbf{D}_2^{\mathrm{CPE}}, \ldots, \mathbf{D}_h^{\mathrm{CPE}}\right)\right) \tag{8}$$

and $\mathbf{D}$ serves as an input and is projected into $\mathbf{K}$ and $\mathbf{V}$ when computing attention. This way can fusion and enhance information communication between different heads in EMSMSA. Thus, the output features of each head contain multiscale information, and then perform the multihead attention calculation. Instead of using (4), we propose

$$\mathrm{head}_i = \mathrm{Attention}\left(\mathbf{X}\mathbf{W}_i^Q, \mathbf{D}\mathbf{W}_i^K, \mathbf{D}\mathbf{W}_i^V\right). \tag{9}$$

Like MSA, EMSMSA performs h self-attention head functions and concatenates the output features of each head.

Note that there is another point we want to explore further. To ensure the efficiency of our EMSMSA, the used dilated convolutions have a stride R to reduce the spatial resolution of $\mathbf{K}$ and $\mathbf{V}$ before the attention operation. In other words, $\mathbf{D} \in \mathbb{R}^{(N/R^2)\times C}$ in (9) has a smaller length than the original $\mathbf{X} \in \mathbb{R}^{N\times C}$ in (4). Therefore, compared with MSA, EMSMSA is more efficient, and the efficiency of EMSMSA will be discussed in Section IV-E2. Besides, EMSMSA has a stronger multiscale information learning capability, which is beneficial to RS scene classification tasks.

D. Cross-Level Attention Module

So far, the obtained $\mathbf{F}_1' \in \mathbb{R}^{H_1\times W_1\times C_1}$, $\mathbf{F}_2' \in \mathbb{R}^{H_2\times W_2\times C_2}$, and $\mathbf{F}_3' \in \mathbb{R}^{H_3\times W_3\times C_3}$ contain rich local information and contextual knowledge within RS scenes. Although self-attention (i.e., (5)) in CIEM can effectively extract contextual information, it restricts to a single-level convolutional feature. To integrate the contributions from different levels, we propose CLAM in this section. It aims to capture long-range correlations between different-level features with various spatial sizes and aggregate multilevel features, such that the contents of RS scenes can be understood comprehensively. Here, the long-range correlation of different-level features can be regarded as the latent dependency information within an image, which is beneficial for constructing a more complete feature representation [11]. Considering the high-level feature ($\mathbf{F}_3'$) contains valuable semantic information, while the low- and middle-level features ($\mathbf{F}_1'$, $\mathbf{F}_2'$) incorporate various shallow messages (such as texture, edge, and color), we embed the shallow messages into semantic information for achieving CLAM's goal. The architecture of CLAM is shown in Fig. 4.

Specifically, to accomplish cross-level attention, we first unify the size of different-level convolutional features. To this end, we apply the average pooling operation and the 1×1 convolutional operation to reduce the spatial size of low-level feature $\mathbf{F}_1'$ and middle-level feature $\mathbf{F}_2'$ and unify their number of channels. After this, $\mathbf{F}_1'$ and $\mathbf{F}_2'$ are transformed into two new features $\mathbf{F}_1^l \in \mathbb{R}^{H_3\times W_3\times C_3}$ and $\mathbf{F}_2^m \in \mathbb{R}^{H_3\times W_3\times C_3}$, respectively, and their sizes are equal to high-level feature $\mathbf{F}_3'$. Second, three learnable parameters $\mathbf{W}^Q$, $\mathbf{W}^K$, $\mathbf{W}^V$ are used to project $\mathbf{F}_1^l$ and $\mathbf{F}_2^m$ into the spaces of $\mathbf{Q}$ and $\mathbf{K}/\mathbf{V}$, respectively. The above steps can be expressed as

$$\mathbf{Q} = \mathbf{F}_1^l\mathbf{W}^Q, \quad \mathbf{K} = \mathbf{F}_2^m\mathbf{W}^K, \quad \mathbf{V} = \mathbf{F}_2^m\mathbf{W}^V. \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

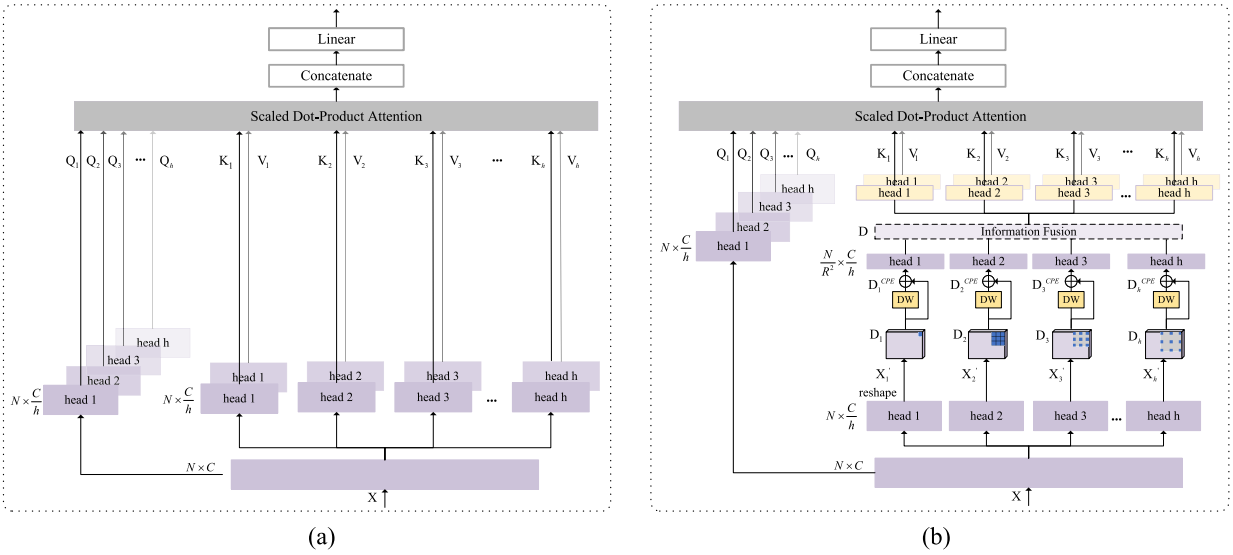


Fig. 3. Illustration of the proposed EMSMSA in EMST. (a) Standard MSA. (b) EMSMSA.

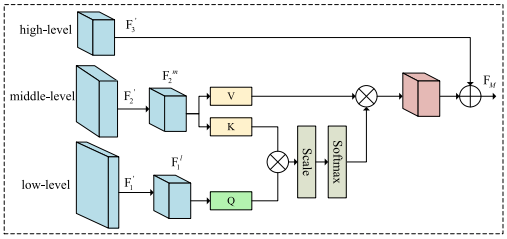


Fig. 4. Illustration of the proposed CLAM.

Third, a dot product and a softmax function are used to compute the correlation weights between $\mathbf{Q}$ and $\mathbf{K}$, which are then multiplied by $\mathbf{V}$ to generate the new features. Finally, the new feature is added to $\mathbf{F}'_3$ for obtaining the final aggregated feature $\mathbf{F}_M \in \mathbb{R}^{H_3 \times W_3 \times C_3}$, which contains multilevel information. In summary, $\mathbf{F}_M$ can be formulated as follows:

$$\mathbf{F}_M = \mathbf{F}'_3 + \text{softmax}\left(\frac{\left(\mathbf{F}_1^l \mathbf{W}^Q\right)\left(\mathbf{F}_2^m \mathbf{W}^K\right)^{\mathrm{T}}}{\sqrt{d_k}}\right)\left(\mathbf{F}_2^m \mathbf{W}^V\right) \quad (11)$$

where $\sqrt{d_k}$ denotes a scaling factor. Note that different from self-attention ((5)) that $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are generated from single-level feature $\mathbf{X}$, CLAM's $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, and the summed features are from different-level features $\mathbf{F}_1^l$, $\mathbf{F}_2^m$, and $\mathbf{F}'_3$, respectively. Therefore, $\mathbf{F}_M$ contains richer and more diverse information of RS scenes.

E. Class Score Fusion Module

In this module, we fuse the multilevel convolutional feature $\mathbf{F}_M$ and global visual feature $\mathbf{g}$ to further improve the performance of RS scene classification. In detail, we first use a global average pooling operation to reduce the spatial size of $\mathbf{F}_M \in \mathbb{R}^{H_3 \times W_3 \times C_3}$ to generate convolutional representation $\mathbf{F}'_M \in \mathbb{R}^{C_3}$. Next, an $L2$-normalization layer is used to normalize convolutional representation and promote stable convergence of the network. Then $\mathbf{F}'_M$ and $\mathbf{g}$ are transformed into classification scores $\mathbf{S}_F \in \mathbb{R}^{C_{\text{class}}}$ and $\mathbf{S}_g \in \mathbb{R}^{C_{\text{class}}}$ through an FC layer, respectively. Finally, to integrate the contributions of two branches, $\mathbf{S}_F$ and $\mathbf{S}_g$ are added to get a fusion classification score $\mathbf{S}$ as

$$\mathbf{S} = \frac{\mathbf{S}_F + \mathbf{S}_g}{2}. \quad (12)$$

Here, the global and local information is fused at the feature level. We use "Score" rather than "Feature" because the dimension of the generated score $\mathbf{S}$ is dynamically decided by the number of categories of the training data. Since $\mathbf{S}$ contains comprehensive information, we use $\mathbf{S}$ as the output of EMTCAL and then a softmax function is used to classify RS scenes. In addition, a cross-entropy loss function is used to optimize the network.

IV. Experiment

A. Dataset Introduction

Three public RS scene datasets are used to evaluate the performance of our EMTCAL, which are the University of California (UC) Merced (UCM) Land Use dataset [53], Aerial Image dataset (AID) [54], and Northwestern Polytechnical University (NWPU)-REmote Sensing Image Scene Classification (RESISC)45 (NWPU) dataset [55].

1) UCM: This dataset was published by the University of California Merced. It contains 2100 RS scene images divided into 21 scene categories. Each category consists of 100 scene images with a size of 256 × 256 pixels, and each pixel has a spatial solution of 0.3 m in the red–green–blue (RGB) color space. Some examples of this dataset are shown in Fig. 5.

2) AID: Wuhan University released this dataset. It comprises 30 scene categories and has 10 000 RS images with a size of 600 × 600 pixels. The number of images in each category varies from 220 to 420, and the spatial resolution changes from about 0.5 m to 8 m per pixel. Some RS scene images of this dataset are shown in Fig. 6.

3) NWPU: This dataset was constructed by Northwestern Polytechnical University and has 31 500 RS scene images. These images are split into 45 scene categories, and each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Examples of the UCM dataset.

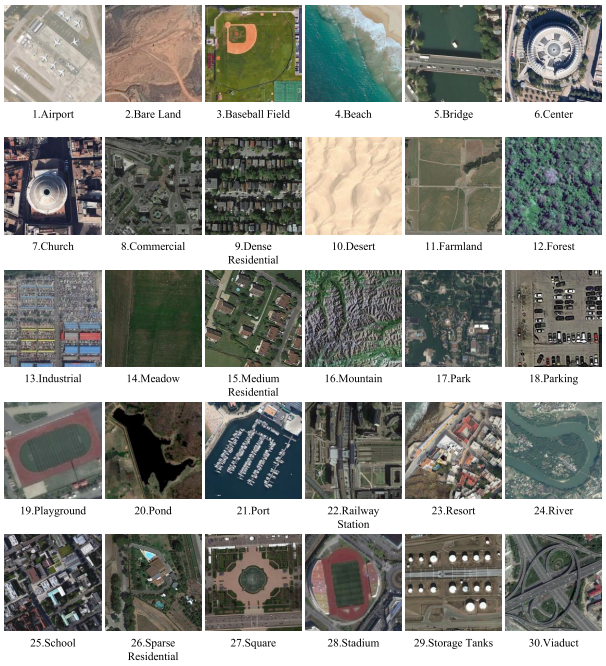


Fig. 6. Examples of the AID dataset.

category contains 700 scene images with 256 × 256 pixels. The spatial resolution varies from 0.2 m to 30 m per pixel. Some image examples are displayed in Fig. 7.

B. Dataset Settings and Evaluation Metrics

We repeat the experiments five times by randomly selecting training/testing samples to obtain reliable experimental results. Then the average results and standard deviations are reported. The training–testing ratios of UCM, AID, and NWPU are set to 50%–50% and 80%–20%, 20%–80% and 50%–50%, and 10%–90% and 20%–80%, respectively. Two widely used evaluation metrics are selected to assess the classification performance, including overall accuracy (OA) and confusion matrix (CM). OA is defined as the number of correctly classified images divided by the total number of testing images, reflecting the overall performance of a classification model. CM is used to analyze the detailed classification errors and confusion degree between different scene classes. In CM, each row/column denotes the true/predicted class.



Fig. 7. Examples of the NWPU dataset.

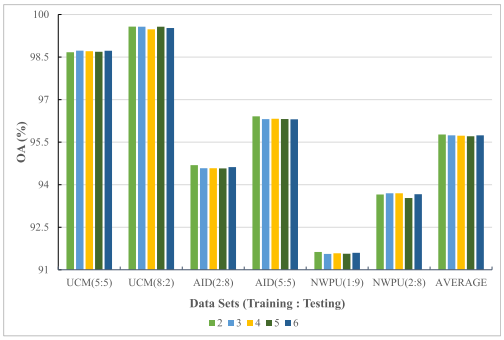


Fig. 8. OA values of our EMTCAL on the UCM, AID, and NWPU datasets under different values of R.

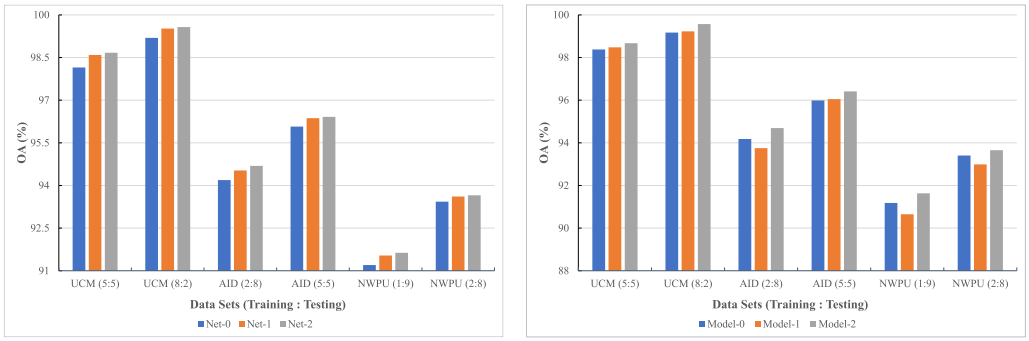
C. Experimental Settings

All the experiments are implemented using Pytorch [56] on a workstation with GeForce RTX 3090 [(35.6 tera floating point operations (TFLOPS)] and 24G Memory. The ResNet34 of EMTCAL is initialized by the pretrained parameters (using the ImageNet dataset [57]), and other parts of our model are initialized randomly. Furthermore, the influence of pretrained parameters is discussed in the supplementary material. We use the Adam optimizer to train our model for 100 epochs. The learning rate is initialized as 0.00003, and a cosine decay learning rate scheduler with a linear warm-up is adopted. The batch size is set to be 32, and all the input scenes are resized into 224 × 224. In addition, the random rotation and horizontal and vertical flipping are used for data augmentation. Considering the size of convolutional feature $\mathbf{C}_3 \in \mathbb{R}^{7\times7\times512}$ and the multiscale information exploration manner in EMST, we set the number of heads as four, and each head corresponds to a dilated convolution with the specific dilation rate (i.e., the kernel's receptive fields are 1 × 1, 3 × 3, 5 × 5, or 7 × 7). By default, we set the hyperparameter stride $R = 2$ in our experiments, and its influence is discussed in Section IV-D.

D. Sensitivity Analysis

This section studies the influence of the free parameter stride R used in our EMTCAL. Here, we conduct experiments on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 9. OA values of different networks on three datasets with different training ratios. (a) Ablation for MFEM, CIEM, and CLAM. (b) Ablation for CSFM.

three datasets using different values of R. Considering the spatial size (i.e., 7×7) of the feature map as mentioned in Section IV-C, we vary the parameter R from 2 to 6 with an interval of 1. The results are summarized in Fig. 8. According to the results, we can find that the performance of our EMTCAL with different parameter values is close, and our EMTCAL achieves the best performance in most cases when $R = 2$. This illustrates that our EMTCAL is not sensitive to the choice of parameter R. Furthermore, we also report the average value of the results on all the datasets for intuitively evaluating the influence of the parameter R in Fig. 8. Similarly, the best classification results of our EMTCAL appear at $R = 2$. Considering the experimental results discussed above, we select $R = 2$ for the following experiments.

E. Ablation Study

To analyze the contributions of different components within our EMTCAL, we construct the ablation experiments on three datasets with different training ratios in this section.

1) Effectiveness of Different Modules: As mentioned in Section III-A, our EMTCAL consists of four modules, including MFEM, CIEM, CLAM, and CSFM. MFEM is used to extract global and convolutional features, CIEM aims at mining contextual information, CLAM can aggregate multilevel features, and CSFM focuses on fusing the global and aggregated features and generating the classification results. In this section, we will study their importance in EMTCAL.

First, we fix CSFM and analyze the contributions of MFEM, CIEM, and CLAM since CSFM has classification function. To this end, we construct the following nets.

1) *Net-0:* MFEM;
2) *Net-1:* MFEM + CIEM + CSFM;
3) *Net-2:* MFEM + CIEM + CLAM + CSFM.

Net-0 is ResNet34, Net-1 is our EMTCAL without CLAM, and Net-2 is our EMTCAL. For Net-0, the global feature learned by ResNet34 is directly used to complete the classification. For Net-1, the concatenate operation is selected to complete the multilayer feature aggregation. Note that the experimental settings for the three networks are set according to the contents discussed in Section IV-C. The results counted on three datasets are shown in Fig. 9(a). From the observation, we can easily find the following points. First, MFEM, CIEM, and CLAM contribute positively to classification performance. Second, the performance of Net-0 is the weakest among the three nets. When introducing the proposed CIEM, Net-1 outperforms Net-0 in all the cases, which illustrates the usefulness of the contextual information for understanding RS scenes. Third, after adding CLAM into Net-1, Net-2 achieves the best classification results, indicating that aggregating and modeling long-range dependencies of multilevel features can further enhance the discrimination of feature representation.

Second, we investigate the contribution of CSFM. As mentioned in Section III-E, CSFM first fuses the multilevel convolutional feature and global visual feature, and then completes the classification. Thus, we construct the following models to inspect the significance of CSFM.

1) *Model-0:* EMTCAL (without multilevel feature);
2) *Model-1:* EMTCAL (without local feature);
3) *Model-2:* EMTCAL.

Model-0 uses the global feature to complete the classification. Model-1 adopts the multilevel convolutional feature to accomplish the classification. Model-2 combines the global and multilevel convolutional features to determine the classification results. Their performance is exhibited in Fig. 9(b). From observing the results, we find that the results of the three models are acceptable. Both the global feature (Model-0) and the multilevel convolutional feature (Model-1) play a positive role in classification. After fusing global and multilevel convolutional features, Model-2 achieves the best performance in all the cases. This demonstrates that integrating the contributions of global and multilevel convolutional features is beneficial to RS scene classification.

2) Effectiveness of EMST: As mentioned in Section III-C, the core component of CIEM is the proposed EMST, which is an EMST architecture to capture contextual information. To study its effectiveness, we compare it with the original transformer architecture (i.e., ViT [44]) and several improved transformer architectures, including Pyramid Vision Transformer (PVT) [27] and Swin transformer (Swin) [46]. For clarity, we only replace the EMST with ViT, PVT, and Swin to construct new network **EMTCAL with ViT**, **EMTCAL with PVT**, and **EMTCAL with Swin**, respectively. Also, for the sake of fairness, there is only one encoder layer in ViT, PVT, and Swin. All the transformer models use four heads in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

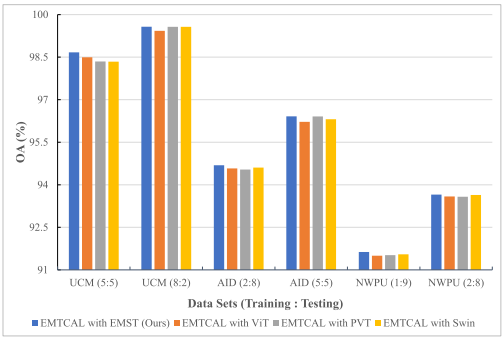


Fig. 10. OA values of EMTCALs with different transformer architectures on three datasets with different training ratios.

TABLE I

GFLOPs OF EMTCALs WITH DIFFERENT TRANSFORMER ARCHITECTURES

Methods	GFLOPs
EMTCAL with PVT	4.25
EMTCAL with Swin	4.27
EMTCAL with ViT	4.27
EMTCAL with EMST (Ours)	**4.23**

the following experiments. Then, we count their classification performance on three datasets with varying training ratios. The experimental settings are the same as the contents mentioned in Section IV-C.

The results of different networks are summarized in Fig. 10. From observing these results, we can find the following conclusions. First, the performance of all the models is acceptable, which indicates the solid contextual information learning ability of the transformer model. Second, the behavior of **EMTCAL with PVT** and **EMTCAL with Swin** is stronger than that of **EMTCAL with ViT** in most cases. This is because ViT is the original architecture, while PVT and Swin are specifically designed to enhance the network feature learning ability for visual tasks. Third, our EMTCAL achieves the best classification performance in all the cases compared with other methods. The reason is that our EMST has both multiscale information learning ability and enhanced contextual information learning ability, which is beneficial to mining the complex contents and their inherent relationships hidden in RS scenes.

Furthermore, to further discover the superiority of the proposed EMST, we also compare its computational complexity with other transformer models using floating point operations (FLOPs). As shown in Table I, giga floating point operations (GFLOPs') value of EMTCAL with EMST is lower than that of other transformer models, which implies the efficiency of EMST. These encouraging experimental results discussed above confirm that our EMST is an effective and efficient transformer model for RS scene classification tasks.

3) Analysis of EMSTs With Different Settings: In this section, we further analyze the impacts of EMSTs with different structures on our EMTCAL.

TABLE II

DETAILED TYPES AND PUBLICATION YEARS OF DIFFERENT METHODS IN OUR EXPERIMENTS. (∗: GLOBAL DEEP-FEATURE-BASED METHODS, •: MULTILEVEL FEATURE-AGGREGATION-BASED METHODS, †: GCN-BASED METHODS, ‡: TRANSFORMER-BASED METHODS)

Methods	Year
GoogLeNet / VGGNet-16 [54] ∗	TGRS2017
VGG-VD16+MSCP+MRA [58] •	TGRS2018
VGG-16-CapsNet [59] ∗	RS2019
SCCov [60] •	TNNLS2019
GBNet+global feature [61] •	TGRS2020
MG-CAP(Sqrt-E) [62] ∗	TIP2020
MIDC-Net_CS [63] ∗	TIP2020
ACR-MLFF [64] •	GRSL2021
ACNet [39] ∗	JSTARS2021
MSA-Network [65] ∗	JSTARS2021
RANet [66] •	JSTARS2021
EFPN-DSE-TDFF [36] •	TGRS2021
DFAGCN [35] † •	TNNLS2021
ViT-B_32 [44] ‡	ICLR2021
T2T-ViT-12 [67] ‡	ICCV2021
PVT-V2-B0 [68] ‡	CVM2022
EMTCAL	Ours

TABLE III

OA VALUES AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE UCM DATASET. THE BEST AND SECOND-BEST VALUES ARE MARKED IN BOLD AND ITALIC

Methods	50% training	80% training
GoogLeNet [54]	92.70±0.60	94.31±0.89
VGGNet-16 [54]	94.14±0.69	95.21±1.20
VGG-16-CapsNet [59]	95.33±0.18	98.81±0.22
SCCov [60]	–	99.05±0.25
VGG-VD16+MSCP+MRA [58]	–	98.40±0.34
GBNet+global feature [61]	97.05±0.19	98.57±0.48
MIDC-Net_CS [63]	95.41±0.40	97.40±0.48
EFPN-DSE-TDFF [36]	96.19±0.13	99.14±0.22
RANet [66]	97.80±0.19	99.27±0.24
DFAGCN [35]	–	98.48±0.42
MG-CAP(Sqrt-E) [62]	–	99.00±0.10
MSA-Network [65]	97.80±0.33	98.96±0.21
ACR-MLFF [64]	*97.99±0.26*	*99.37±0.15*
ViT-B_32 [44]	97.83±0.16	98.95±0.24
T2T-ViT-12 [67]	97.79±0.20	99.10±0.32
PVT-V2-B0 [68]	97.94±0.44	98.86±0.38
EMTCAL (ours)	**98.67±0.16**	**99.57±0.28**

First, we study the influence of EMSTs with different dilation rates on EMTCAL. As mentioned in Section IV-C, the dilation rates are equal to {1, 3, 5, 7} for multilevel features $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_3$. To study the influence of the dilation rates on our EMTCAL, we construct a new network named EMTCAL-D. Unlike EMTCAL, EMTCAL-D applies various dilation rates to different feature maps, and the dilation rates are decided according to the sizes of feature maps $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_3$. In particular, the dilation rates of EMST in EMTCAL-D are {1, 9, 17, 25}, {1, 5, 9, 13}, and {1, 3, 5, 7} for low-, middle-, and high-level convolutional feature maps, respectively. For fairness, the experimental settings of EMTCAL-D are the same as the contents discussed in Section IV-C. The results of two networks counted on three datasets with different training sets are exhibited in Fig. 11(a). From the observation of the bars,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

OA VALUES AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE AID DATASET. THE BEST AND SECOND-BEST VALUES ARE MARKED IN BOLD AND ITALIC

Methods	20% training	50% training
GoogLeNet [54]	83.44±0.40	86.39±0.55
VGGNet-16 [54]	86.59±0.29	89.64±0.36
VGG-16-CapsNet [59]	91.63±0.19	94.74±0.17
SCCov [60]	93.12±0.25	96.10±0.16
VGG-VD16+MSCP+MRA [58]	92.21±0.17	95.56±0.18
GBNet+global feature [61]	92.20±0.23	95.48±0.12
MIDC-Net_CS [63]	88.51±0.41	92.95±0.17
EFPN-DSE-TDFF [36]	94.02±0.21	94.50±0.30
RANet [66]	92.71±0.14	95.31±0.37
ACNet [39]	93.33±0.29	95.38±0.29
DFAGCN [35]	–	94.88±0.22
MG-CAP(Sqrt-E) [62]	93.34±0.18	96.12±0.12
MSA-Network [65]	93.53±0.21	96.01±0.43
ACR-MLFF [64]	92.73±0.12	95.06±0.33
ViT-B_32 [44]	93.74±0.27	95.84±0.29
T2T-ViT-12 [67]	*94.39±0.22*	*96.29±0.24*
PVT-V2-B0 [68]	93.52±0.35	96.27±0.14
EMTCAL (ours)	**94.69±0.14**	**96.41±0.23**

TABLE V

OA VALUES AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE NWPU DATASET. THE BEST AND SECOND-BEST VALUES ARE MARKED IN BOLD AND ITALIC

Methods	10% training	20% training
GoogLeNet [55]	76.19±0.38	78.48±0.26
VGGNet-16 [55]	76.47±0.18	79.79±0.15
VGG-16-CapsNet [59]	85.08±0.13	89.18±0.14
SCCov [60]	89.30±0.35	92.10±0.25
VGG-VD16+MSCP+MRA [58]	88.07±0.18	90.81±0.13
MIDC-Net_CS [63]	86.12±0.29	87.99±0.18
ACNet [39]	*91.09±0.13*	92.42±0.16
DFAGCN [35]	–	89.29±0.28
MG-CAP(Sqrt-E) [62]	90.83±0.12	92.95±0.13
MSA-Network [65]	90.38±0.17	*93.52±0.21*
ACR-MLFF [64]	90.01±0.33	92.45±0.20
ViT-B_32 [44]	90.05±0.29	92.61±0.14
T2T-ViT-12 [67]	90.62±0.18	93.19±0.10
PVT-V2-B0 [68]	89.72±0.16	92.95±0.09
EMTCAL (ours)	**91.63±0.19**	**93.65±0.12**

we can find the behavior of EMTCAL-D is stronger than that of EMTCAL on UCM and NWPU, with the training ratios of 50% and 20%, respectively. In other cases, EMTCAL outperforms EMTCAL-D. Fortunately, their performance gap is small, illustrating that the proposed EMTCAL is not sensitive to the different dilation rates. Thus, we suggest that the dilation rates can be set equally for various feature maps for convenience.

Second, we study the influence of the numbers of EMST layers on our EMTCAL. Here, we use two EMSTs to construct a new network EMTCAL-L. The performance of EMTCAL and EMTCAL-L on different datasets is shown in Fig. 11(b). From observing the results, we find that EMTCAL-L outperforms EMTCAL in most cases on three datasets with different training ratios, which illustrates that the more the EMSTs, the stronger the EMTCAL. The reason is that more EMSTs can obtain richer contextual information. However, as the number of EMST layers increases, the computational complexity of the model becomes larger. Considering the trade-off between the performance and computation complexity, we use one EMST layer in this article.

Apart from the above ablation studies, we also discuss whether the transformer part of our EMTCAL (i.e., EMST) can achieve good classification results or not in the supplementary material.

F. Comparisons of State-of-the-Art Models

To comprehensively evaluate the classification performance of the proposed EMTCAL, we compare our method with some state-of-the-art models, including global deep-feature-based methods, multilevel feature-aggregation-based methods, GCN-based methods, and transformer-based methods. The detailed information of those compared models is summarized in Table II. The transformer-based methods are accomplished by ourselves, and the experimental settings are the same as ours. Meanwhile, the results of other methods are from the original literature.

1) Results on UCM Dataset: The classification results of different methods are summarized in Table III. Compared with the conventional CNN methods (e.g., VGGNet-16), multilevel feature aggregation-based methods (e.g., VGG-VD16 + multilayer stacked covariance pooling (MSCP) + multiresolution analysis (MRA), ACR-MLFF) achieve better classification performance. This is because the features produced by the aggregation strategy are more discriminative. The performance of the transformer-based models is stronger than that of the conventional CNN models, which confirms the potential of the transformer model in mining global contextual information. Although the performance of the above methods is good, our EMTCAL outperforms them in all the cases. Compared with the best feature-aggregation-based model (i.e., ACR-MLFF) and transformer-based model (i.e., PVT-V2-B0), the improvements in OA obtained by EMTCAL are 0.68% (over ACR-MLFF) and 0.73% (over PVT-V2-B0) when the training ratio is 50%. When the training ratio equals 80%, the improvements are 0.20% (over ACR-MLFF) and 0.47% (over T2T-ViT-12). The reason behind this can be summarized as follows. On one hand, our EMTCAL can aggregate local information and their contextual relationships within RS scenes to generate discriminative feature representation. On the other hand, EMTCAL can effectively extract single-level multiscale contextual information and cross-level long-range dependencies of different-level features so that the contents of RS scenes can be fully interpreted.

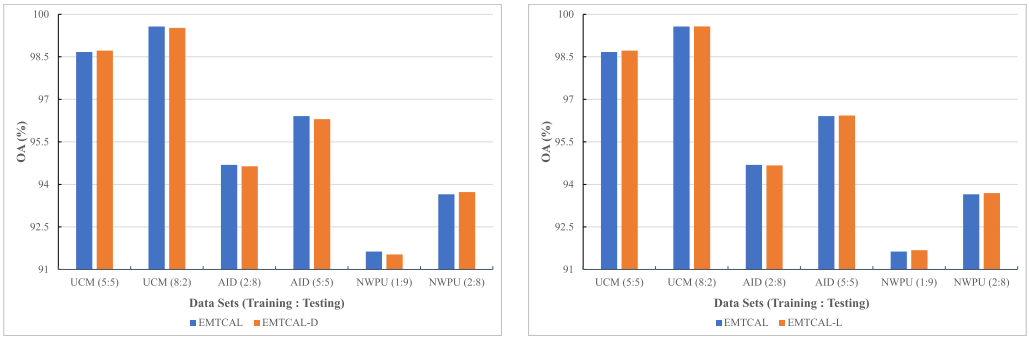
Apart from OA, we also report CMs of our EMTCAL under different training ratios in Fig. 12. From observing CMs, we can find that most scene categories are correctly distinguished. Take Fig. 12(b) as an example, the confusion only appeared in "Dense Residential" and "Intersection." The encouraging results discussed above prove that our EMTCAL is effective for the UCM dataset.

2) Results on AID Dataset: Table IV reports the classification results of different methods on the AID dataset. Similar to UCM, we can find that our EMTCAL achieves the best performance in all the cases. When 20% scenes are used for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

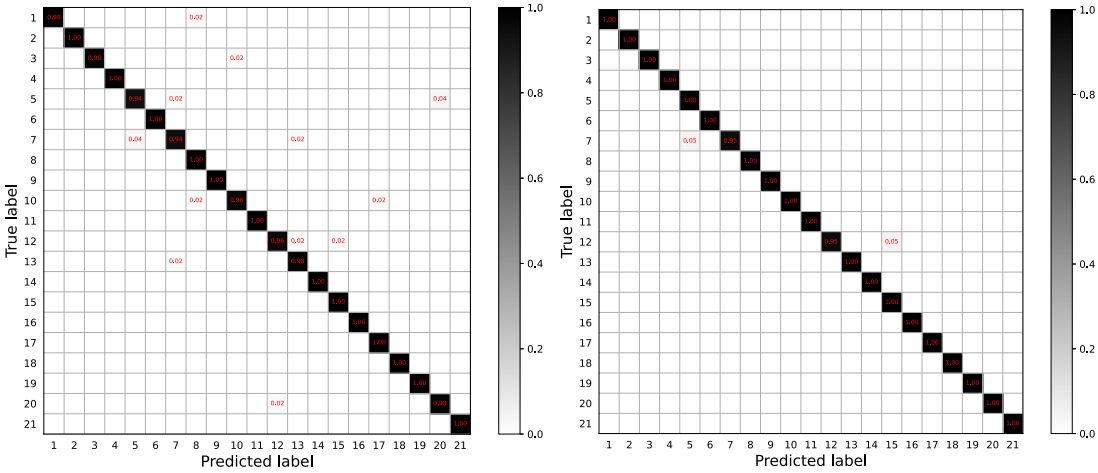
TABLE VI
TRAINING AND TESTING TIME OF DIFFERENT METHODS ON THREE DATASETS UNDER DIFFERENT TRAINING RATIOS

	GBNet [61]		EFPN [36]		ACNet [39]		EMTCAL (ours)	
	Train.(m)	Test.(s)	Train.(m)	Test.(s)	Train.(m)	Test.(s)	Train.(m)	Test.(s)
UCM (50% training)	9.25	2.15	7.49	1.85	12.57	2.88	5.33	1.40
UCM (80% training)	14.40	1.13	11.49	1.02	19.79	1.41	8.33	0.98
AID (20% training)	17.88	24.54	13.96	24.05	24.41	24.52	11.61	23.53
AID (50% training)	42.33	15.07	32.96	15.12	58.55	16.07	26.83	15.04
NWPU (10% training)	26.32	46.77	20.80	37.50	36.64	67.69	15.41	27.15
NWPU (20% training)	51.35	41.22	40.73	33.23	72.63	61.52	29.53	23.68



(a) (b)

Fig. 11. OA values of EMTCALs with different EMSTs, including different (a) dilation rates and (b) layer numbers on three datasets with different training ratios.



(a) (b)

Fig. 12. Confusion matrices of EMTCAL under different training ratios on the UCM dataset. The scene category corresponding to each number can be found in Fig. 5. (a) 50%. (b) 80%.

training, the OA of our EMTCAL achieves 94.69%, which is 0.30% higher than the second-best method (T2T-ViT-12). When 50% scenes are used for training, our EMTCAL achieves 96.41%, which is 0.12% higher than the second-best method, i.e., T2T-ViT-12. An interesting observation is that the classification behavior of the RS-oriented CNN-based networks (ACNet and DFAGCN) is significantly stronger than that of the conventional CNN models (GoogLeNet and VGGNet-16). This implies the importance of the consideration of RS scenes' specific characteristics. Compared with the general-purpose CNN models, ACNet and DFAGCN introduce attention mechanisms and GCN into the feature learning process, which is conducive to exploring the complex contents within RS scenes. The observation discussed above is confirmed by our EMTCAL's encouraging results again. Due to the multiscale learning strategy and the contextual relation mining mechanism, RS scenes' sufficient and diverse land covers can be fully captured. These positive results discussed above verify the usefulness of our method.

Besides, CMs of our EMTCAL under different training ratios are exhibited in Fig. 13. When the training rate is 20% and 50%, 24 and 28 of the 30 categories have classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

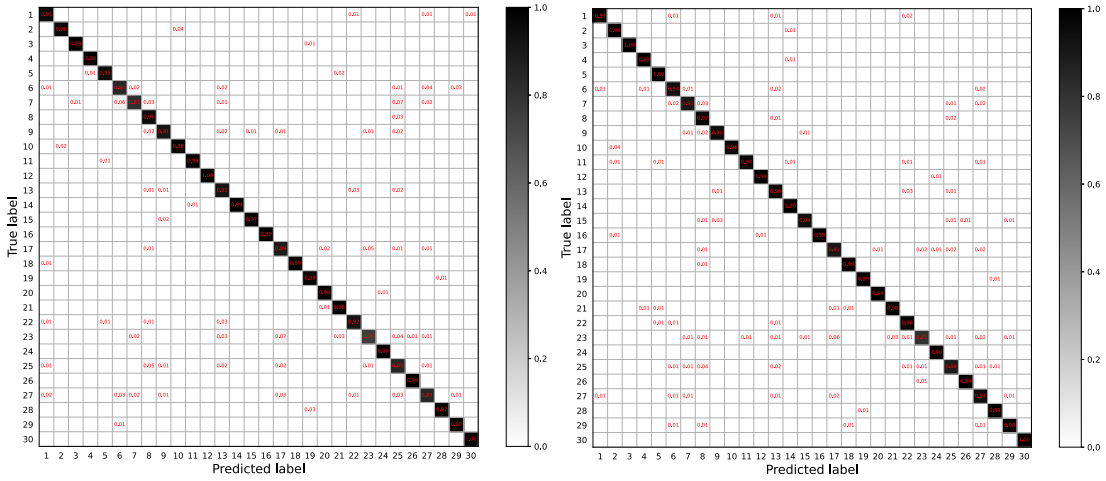


Fig. 13. Confusion matrices of EMTCAL under different training ratios on the AID dataset. The scene category corresponding to each number can be found in Fig. 6. (a) 20%. (b) 50%.

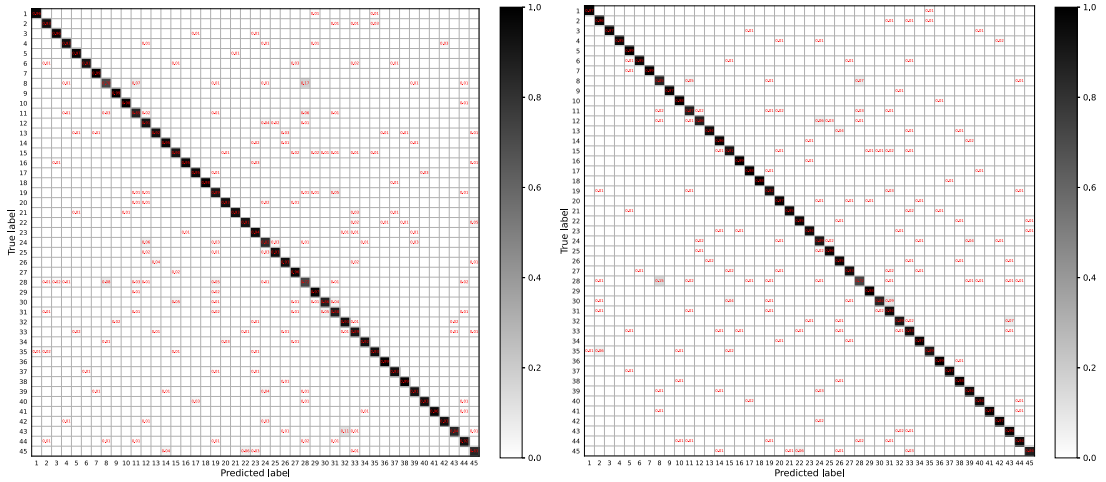


Fig. 14. Confusion matrices of EMTCAL under different training ratios on the NWPU dataset. The scene category corresponding to each number can be found in Fig. 7. (a) 10%. (b) 20%.

accuracies over 90%, respectively. Nevertheless, the performance of EMTCAL is not as good as what we expect in some scenes, such as "Resort" and "Parking." This is because of the high interclass similarity. Addressing this limitation and further improving EMTCAL's performance are our future works.

3) Results on NWPU Dataset: The results of different methods are summarized in Table V, in which we can find that the performance of our EMTCAL is best. Compared with other methods, the enhancements in OA obtained by EMTCAL under the training ratio of 10% are 15.44% (over GoogLeNet), 15.16% (over VGGNet-16), 6.55% (over VGG-16-CapsNet), 2.33% [over skip-connected covariance network (SCCov)], 3.56% (over VGG-VD16 + MSCP + MRA), 5.51% [over multiple instance densely-connected ConvNet (MIDC-Net_CS)], 0.54% (over ACNet), 0.80% [over multi-granularity canonical appearance pooling (MG-CAP)(Sqrt-E)], 1.25% (over MSA-Network), 1.62% (over ACR-MLFF), 1.58% (over ViT-B_32), 1.01% (over T2T-ViT-12), and 1.91% (over PVT-V2-B0). The enhancements obtained by EMTCAL under the training ratio of 20% are 15.17% (over GoogLeNet), 13.86% (over VGGNet-16), 4.47% (over VGG-16-CapsNet), 1.55% (over SCCov), 2.84% (over VGG-VD16 + MSCP + MRA), 5.66% (over MIDC-Net_CS), 1.23% (over ACNet), 4.36% (over DFAGCN), 0.70% [over MG-CAP(Sqrt-E)], 0.13% (over MSA-Network), 1.20% (over ACR-MLFF), 1.04% (over ViT-B_32), 0.46% (over T2T-ViT-12), and 0.70% (over PVT-V2-B0). These promising results verify the superiority of our method for the RS scene classification task.

Furthermore, CMs of our EMTCAL on the NWPU dataset are shown in Fig. 14. According to Fig. 14(a), when 10% scenes are used for training, 35 of the 45 categories have classification accuracies over 90%. As shown in Fig. 14(b), 36 of the 45 categories have classification accuracies over 90% when 20% scenes are used for training. These positive results prove the effectiveness of our method again. However,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

EMTCAL performs weakly in some scene categories, such as "Church" and "Place" due to the similar semantic distribution among these scenes.

Besides the above comparisons, we also study the performance of the proposed EMTCAL under limited supervision, and the EMTCAL's behavior changes with the variation in the number of training samples. The related discussion can be found in the supplementary material.

G. Time Costs

As discussed in Section IV-E, we have compared the computational complexity of our EMTCAL with some recent transformer architectures and verified the efficiency of our EMTCAL. In this section, we further study its computational complexity. To this end, the training and inference time costs of EMTCAL are counted based on different datasets and various training sets. Also, the time costs of gated bidirectional network (GBNet) [61], EFPN [36], and ACNet [39] are counted for reference. The results are summarized in Table VI. It is easy to find that the time costs of our EMTCAL are the lowest. These positive results confirm the superiority of our method for RS scene classification tasks again.

V. Conclusion

This article proposes a new RS scene classification method named EMTCAL. It mainly consists of four modules, i.e., MFEM, CIEM, CLAM, and CSFM. In MFEM, we adopt ResNet34 to learn the multilevel convolutional and global features from RS scenes. Then, CIEM uses the developed EMST to further analyze the convolutional feature maps from each level, respectively. Such that not only can the numerous contents with diverse scale information hidden in RS scenes be captured but also their inherent contextual relationships can be explored. Next, CLAM integrates the rich knowledge from different levels in a self-attention manner. In this way, the important information corresponding to each level can be kept and the long-range dependencies between multilevel features can be established, which ensures the discrimination of the obtained representations. Finally, CSFM applies a simple fusion strategy to combine the contributions of global and aggregated multilevel features. The promising experimental results counted on three widely used datasets illustrate that our EMTCAL is helpful to the RS scene classification tasks.

Although the proposed EMTCAL achieves encouraging classification results, there is still room for improvement. First, we use ResNet34 to directly learn the multilevel features from RS scenes. Although it is feasible, the general-purpose CNN may ignore the specific characteristics of RS scenes. One of our future works is to construct RS-oriented CNN to explore the multilevel features within RS scenes. Second, to avoid increasing the model's complexity, the fusion scheme used in CSFM is simple, which would limit our model's performance. Selecting or developing more supportive fusion ways is another future work. Besides, we plan to introduce self-supervision learning [69] into our model to enhance its capacity in future work.

References

[1] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[2] X.-Y. Tong *et al.*, "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[3] X. Tang *et al.*, "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[4] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[5] U. Maulik and D. Chakraborty, "Remote sensing image classification: A survey of support-vector-machine-based advanced techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 33–52, Mar. 2017.

[6] Y. Gu, Y. Wang, and Y. Li, "A survey on deep learning-driven remote sensing image scene understanding: Scene classification, scene retrieval and scene-guided object detection," *Appl. Sci.*, vol. 9, no. 10, p. 2110, 2019.

[7] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 886–893.

[8] Y. Yang and S. Newsam, "Comparing SIFT descriptors and Gabor texture features for classification of remote sensed imagery," in *Proc. 15th IEEE Int. Conf. Image Process.*, Oct. 2008, pp. 1852–1855.

[9] C. Liu, J. Ma, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Deep hash learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3420–3443, Apr. 2021.

[10] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[11] M. Li, L. Lei, Y. Sun, X. Li, and G. Kuang, "A cross-layer nonlocal network for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[12] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[13] X. Tang *et al.*, "Meta-hashing for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2021.

[14] G. Cheng *et al.*, "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[15] X. Tang, W. Lin, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Class-level prototype guided multiscale feature learning for remote sensing scene classification with limited labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[16] X. Tang, X. Zhang, F. Liu, and L. Jiao, "Unsupervised deep feature learning for remote sensing image retrieval," *Remote Sens.*, vol. 10, no. 8, p. 1243, Aug. 2018.

[17] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[18] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2021.

[19] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[20] Z. Peng *et al.*, "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 367–376.

[21] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[22] T. Lin, Y. Wang, X. Liu, and X. Qiu, "A survey of transformers," 2021, *arXiv:2106.04554*.

[23] K. Han *et al.*, "A survey on vision transformer," 2020, *arXiv:2012.12556*.

[24] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, Dec. 2021, doi: 10.1145/3505244.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.

[25] K. Lin, L. Wang, and Z. Liu, "End-to-end human pose and mesh reconstruction with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1954–1963.
[26] W. Xu, Y. Xu, T. Chang, and Z. Tu, "Co-scale conv-attentional image transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9981–9990.
[27] W. Wang *et al.*, "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 568–578.
[28] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.
[29] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
[30] E. Choi *et al.*, "Learning the graphical structure of electronic health records with graph convolutional transformer," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 1, pp. 606–613.
[31] W. Li *et al.*, "Classification of high-spatial-resolution remote sensing scenes method using transfer learning and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1986–1995, 2020.
[32] J. Xie, N. He, L. Fang, and A. Plaza, "Scale-free convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6916–6928, Sep. 2019.
[33] Y. Zhang, X. Zheng, Y. Yuan, and X. Lu, "Attribute-cooperated convolutional neural network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8358–8371, Dec. 2020.
[34] X. Lu, H. Sun, and X. Zheng, "A feature aggregation convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7894–7906, Oct. 2019.
[35] K. Xu, H. Huang, P. Deng, and Y. Li, "Deep feature aggregation framework driven by graph convolutional network for scene classification in remote sensing," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 15, 2021, doi: 10.1109/TNNLS.2021.3071369.
[36] X. Wang, S. Wang, C. Ning, and H. Zhou, "Enhanced feature pyramid network with deep semantic embedding for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7918–7932, Sep. 2021.
[37] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.
[38] J. Wang, Y. Zhong, Z. Zheng, A. Ma, and L. Zhang, "RSNet: The search for remote sensing deep neural networks in recognition tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2520–2534, Mar. 2021.
[39] X. Tang, Q. Ma, X. Zhang, F. Liu, J. Ma, and L. Jiao, "Attention consistent network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2030–2045, 2021.
[40] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2021.
[41] J. Lu *et al.*, "SOFT: SoftMax-free transformer with linear complexity," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 21297–21309.
[42] F. Zhu, Y. Zhu, L. Zhang, C. Wu, Y. Fu, and M. Li, "A unified efficient pyramid transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2667–2677.
[43] D.-J. Chen, H.-Y. Hsieh, and T.-L. Liu, "Adaptive image transformer for one-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 12247–12256.
[44] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[45] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 357–366.
[46] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 10012–10022.
[47] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.
[48] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, 2021.
[49] J. Ma, M. Li, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Homo–heterogenous transformer learning framework for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2223–2239, 2022.
[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[51] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.
[52] X. Chu *et al.*, "Conditional positional encodings for vision transformers," 2021, *arXiv:2102.10882*.
[53] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, 2010, pp. 270–279.
[54] G. S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[55] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[56] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.
[57] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1–9.
[58] N. He, L. Fang, S. Li, A. Plaza, and J. Plaza, "Remote sensing scene classification using multilayer stacked covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 6899–6910, Dec. 2018.
[59] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, 2019.
[60] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.
[61] H. Sun, S. Li, X. Zheng, and X. Lu, "Remote sensing scene classification by gated bidirectional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 82–96, Jan. 2020.
[62] S. Wang, Y. Guan, and L. Shao, "Multi-granularity canonical appearance pooling for remote sensing scene classification," *IEEE Trans. Image Process.*, vol. 29, pp. 5396–5407, 2020.
[63] Q. Bi, K. Qin, Z. Li, H. Zhang, K. Xu, and G.-S. Xia, "A multiple-instance densely-connected ConvNet for aerial scene classification," *IEEE Trans. Image Process.*, vol. 29, pp. 4911–4926, 2020.
[64] X. Wang, L. Duan, A. Shi, and H. Zhou, "Multilevel feature fusion networks with adaptive channel dimensionality reduction for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.
[65] G. Zhang *et al.*, "A multiscale attention network for remote sensing scene images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9530–9545, 2021.
[66] X. Wang, L. Duan, C. Ning, and H. Zhou, "Relation-attention networks for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 422–439, 2021.
[67] L. Yuan *et al.*, "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 558–567.
[68] W. Wang *et al.*, "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, pp. 415–424, Mar. 2022.
[69] X. Liu *et al.*, "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, early access, Jun. 22, 2021, doi: 10.1109/TKDE.2021.3090866.



Xu Tang (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electronic circuit and system from Xidian University, Xi'an, China, in 2007, 2010, and 2017, respectively.

From 2015 to 2016, he was a Joint Ph.D. Student along with Prof. W. J. Emery with the University of Colorado at Boulder, Boulder, CO, USA. He is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include remote sensing image content-based retrieval and reranking, hyperspectral image processing, remote sensing scene classification, object detection, etc. For more details, refer to: https://web.xidian.edu.cn/tangxu/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.



Mingteng Li received the B.E. degree in computer science and technology from the Qingdao University of Science and Technology, Qingdao, China, in 2020. He is pursuing the M.S. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include machine learning, vision transformer, and remote sensing image processing.



Jingjing Ma (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2004 and 2012, respectively.

She is currently an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. Her research interests include computational intelligence and image understanding.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees from the School of Computer Science, Xidian University, Xi'an, China, in 1999 and 2003, respectively, and the Ph.D. degree from the School of Electronic Engineering, Xidian University, in 2006.

She is a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University. From January 2015 to March 2016, she was a Visiting Scientist with the Computer Science and Artificial Intelligence Laboratory, Massachusetts Institute of Technology, Cambridge, MA, USA. Her research interests include pattern recognition, machine learning, and remote sensing image analysis and understanding.



Fang Liu (Member, IEEE) was born in China, in 1990. She received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is currently an Associate Professor with the Nanjing University of Science and Technology, Nanjing, China. Her research interests include deep learning, object detection, polarimetric SAR image classification, and change detection.



Licheng Jiao (Fellow, IEEE) received the B.S. degree in high voltage from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees in electronic engineering from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China. He has authored or coauthored more than 200 scientific articles. His research interests include signal and image processing, nonlinear circuits and systems theory, wavelet theory, natural computation, and intelligent information processing.

Dr. Jiao is also a member of the IEEE Xian Section Executive Committee and an Executive Committee Member of the Chinese Association of Artificial Intelligence. He is the Chairman of the Awards and Recognition Committee.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:27:38 UTC from IEEE Xplore. Restrictions apply.